

LESSON 132 (0.5) Ground (1.5) Cross-Country Dual (0.3) Instrument

Lesson Objective

The purpose of this lesson is to introduce cross-country flight procedures including navigation by means of pilotage and dead reckoning. The applicant will practice calculating ground speed and fuel burn by using actual time enroute (ATE) and comparing the results with their preflight planning.

Pre-Brief

- Review Maneuvers and Procedures to be Performed in Flight
- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Student should have completed navigation logs for the planned route with minimal errors
- During this lesson quiz the student on the following and provide instruction where needed.
- Student questions.

References

- Aviation Weather Handbook ([FAA-H-8083-28A](#))
- Advisory Circulars.
 - Pilot's Guide to a Preflight Briefing (AC91-92)
- Aeronautical Information Manual
 - Chapter 1: Air Navigation
 - Chapter 3: Airspace
 - Chapter 4: Air Traffic Control
 - Section 3: Airport Operations
 - Chapter 6: Emergency Procedures
 - Chapter 7: Safety of Flight
- Pilot's Information Manual
 - Section 4: Normal Procedures
 - Section 5: Performance

Instructor Guidance

- Thoroughly review the students navigation logs to ensure they were completed correctly with minimal mistakes.
 - Verify altitudes are correct based off of magnetic course.
 - Ensure the student has noted TOC and TOD.
 - Information for the fuel management table should match the information on the navlog.
 - Ensure the flight plan portion has all required information and is correctly filled out.
- Do an in-depth briefing to discuss the all components of 14 CFR Part 91.103
 - Verify the fuel information on the TOLD card matches the information on the navlog.
 - Ensure the student has received a weather briefing.

- Provide the student with insight on how to correctly fill out the navigation logs at each point and use pilotage and dead-reckoning.
- File the flight plans with FSS.
- Brief the student on how to:
 - Open the flight plan.
 - Pick up flight following.

Lesson Content

Checklist Procedures

- Have the student run through all checklists.
 - Introduce requirement for inputting flight plan into GPS during checklist item "PFD...Configure for departure" and verify planned altitude on G5.
- Incorporate flows for memory items.
 - Discuss requirement for Cruise Checklist when leveled off in flight.
- Student should be able to perform all briefings.

Preflight Inspection

- Ensure that the student conducted a thorough preflight inspection.

Operation of the Communications and Navigation Units

- Ensure that the student can proficiently use COMS and NAV units.
 - Student should be able to correctly tune and ID a VOR.
 - Have the student put the appropriate approach frequency in to pick up flight following.
- Confirm the student has correctly input the flight plan for the route planned and verify it is shown correctly on the map page.

Engine Starting

- Correctly determine and use appropriate start procedure.
- Incorporate the use of flows.

Radio Communication Procedures and ATC Phraseology

- The student should be able to conduct radio calls in towered airport.
- The student should be able to conduct radio calls in the practice area.
- Introduce phrasology for opening flight plan with FSS:
 - Opening the flight plan:
 - Pilot: "St-Petersburg radio, N612FT, transmitting and receiving on 122.2".
 - FSS: "N612FT, St. Petersburg radio, go ahead".
 - Pilot: "St-Petersburg radio, N612FT would like to open our flight plan from KMLB to KOBE".
 - FSS: "N612FT, St. Petersburg radio, flight plan is activated".
 - Closing the flight plan:
 - Pilot: "St-Petersburg radio, N612FT, transmitting and receiving on 122.2"

- FSS: "N612FT, St. Petersburg radio, go ahead".
- Pilot: "St-Petersburg radio, N612FT would like to close our flight plan from KMLB to KOBE".
- FSS: "N612FT, St. Petersburg radio, flight plan is terminated".
- Introduce phraseology for VFR flight following:
 - Picking up:
 - Pilot: "Orlando approach, TCF612 6 miles South-West of Melbourne, requesting VFR flight following to KOBE, Altitude 3000ft, Type Papa 28 Alpha"
 - ATC: "TCF612, Orlando Approach, squawk 1234"
 - Pilot: " Squawk 1234, TCF612"
 - ATC: "TCF612, Orlando Approach, radar contact 6 miles South-West of Melbourne, Altimeter 30.01"
 - Cancelling:
 - Pilot: "Orlando approach, TCF612, requesting to cancel VFR Flight following"
 - ATC: "TCF612, Orlando approach, radar services terminated, squawk VFR, frequency change approved"
 - Pilot: "TCF612, radar services terminated, squawk VFR, frequency change approved"

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command's responsibility to avoid traffic.
 - Flight following can assist with traffic alerts but PIC is still responsible to see and avoid.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Taxiing

- The student should be able to taxi on the centerline at appropriate speeds on the ramp and taxiways without instructor guidance
- Ensure the correct use of wind correction throughout taxi.
- Verify the student has airport diagrams for new airports and ensure they brief all taxi instructions.
 - Briefing taxi routes is always required but should be emphasized at unfamiliar airports.

Pilotage and Dead Reckoning:

- Pilotage: navigation by reference to landmarks or checkpoints.
 - A method of navigation that can be used on any course with adequate checkpoints but is more commonly used with dead reckoning and VHF radio navigation.
 - The checkpoint used should be prominent features common to the area of flight.
 - Roads, rivers, railroad tracks, lakes, and power lines.
 - If possible, select features that will make useful boundaries on each side of the course.
 - Keep from drifting too far off course by referring to and not crossing selected points.
 - Never place complete reliance on any single checkpoint, choose ample checkpoints.
 - If the checkpoint is not in sight, do not continue blindly.
- Dead reckoning: navigation solely by computation based on time, airspeed, distance, and direction.

- The products derived from these, when adjusted by wind speed and velocity, are heading and ground speed (GS).
- The predicted heading will guide the airplane along the intended path and the GS will establish the time to arrive at each checkpoint and destination.

Navigation Systems and Radar Services

- Navigation Systems
 - Verify all GPS databases are up to date on the G430 or G650.
 - Student should be able to correctly input all required frequencies and navigate correctly using the nearest page.
 - Provide assistance to student to input flight plan and verify the CDI is on the correct mode to display either GPS or VOR information on the G5 or PFD of Avidyne.
 - Student should be able to correctly tune and ID a VOR.
- Radar Services
 - Understand primary radar (range and azimuth only) and secondary radar (transponder-based — Mode A/C/S).
 - Transponder codes:
 - 1200 – VFR, 7500 – Hijack, 7600 – Radio Failure, 7700 – Emergency.
 - When working with ATC radar:
 - Listen for assigned discrete transponder code.
 - Respond to radar vectors using standard-rate turns and precise altitude control.
 - Read back headings and altitudes to ensure clarity.
 - If transitioned to a different ATC sector:
 - ATC will provide a frequency to change to. Check in by transmitting “Palm Beach Approach, TCF612, 3000ft”
 - If given a heading to follow, state the heading you are on.
 - If unable to hear ATIS/AWOS when attempting to pick it up or only operating using one radio, request a frequency change with ATC; ATC may give you a time frame to check back in with them.

Flight Service Station Flight Plan

- During pre-brief, review the student’s flight plan on the back of their navlog; correct any mistakes.
- Have the student file the flight plan for both directions of flight via 1800wxbrief.com, by calling 1-800-WXBRIEF, through Foreflight, or through SkyVector. Monitor to ensure accuracy.
- Suggest that the student signs up for EasyActivate/EasyClose so they may open and close flight plans through their phone or email.
- To open or close the flight plan in their air, tune to the correct flight service frequency and refer to the section on **Radio Communication Procedures and ATC Phraseology** for what to say to the FSS.
 - If already on Flight Following, request a frequency change with ATC to open or close plan.
- DO NOT FORGET TO CLOSE YOUR FLIGHT PLAN!

- Flight service will attempt to contact you if the flight plan is still open.
- After 30 minutes of no response, Flight Service may send search and rescue to attempt to locate your aircraft.
- Flight Service can also provide in-flight weather briefings if unforecasted weather appears along your route of flight.

Navigation Log

- Departure
 - Ensure flight plan, radios, and NAVaids are correctly configured.
 - Complete Runway Entry Check:
 - Note TIME OFF on navlog and fill out ETA of the first point.
 - Follow ATC instruction to proceed on course.
- Cruise
 - When leveled off at correct altitude, complete Cruise Check
 - Ensure the student knows the proper leaning procedures for Best Power vs. Best Economy
 - Make sure student is switching tanks when required
 - Evaluate CRM and division of attention
 - Ensure traffic scanning when filling out navlog
 - Pilotage and Dead Reckoning
 - Help student verify checkpoints
 - Ensure course is intercepted, bug and follow planned heading
 - Record actual Ground speed
 - ETAs, ATAs, ETEs, ATEs all recorded on navlog at each checkpoint
 - If incorrect or inaccurate, revise using E6B
 - Fill out fuel burn
 - Radio Communication
 - Opening Flight Plan
 - Picking up Flight Following
 - Set required frequencies when in between points
- Arrival
 - Pick up ATIS/AWOS of destination airport when able
 - Start descent at calculated TOD
 - Do Descent and Arrival checklist prior to descent.
 - Thoroughly brief arrival for pattern entry, expected runway, and runway lengths.
 - Monitor frequency to determine runways in use.
 - If non-towered, make radio call 10 miles from airport to report position and intentions.
 - Radio Communication
 - Announce intent to descend to ATC.
 - Contact Flight Service to close flight plan.
 - Complete navlog when safe to do so.

Emergency Approach and Landing / Systems and Equipment Malfunction

- Systems and Equipment Malfunction
 - Fail the MFD/PFD. Take away the moving map and all nav aids.
 - Ensure student completes correct checklists.
 - Pilotage as main method of navigation.
 - Continue to fill out navlog and verify checkpoints.
 - Communicate with ATC if required to declare emergency.
- Emergency Procedures (Engine)
 - Give student partial power loss/Engine roughness.
 - Evaluate decision making
 - Provide student insight on circling or going direct to airport depending on severity of issue.
 - Ensure flows are used for emergency checklist but have the student verify.
 - AVIATE, NAVIGATE, COMMUNICATE!
 - Complete navlog and diversion log if time permits.
 - Use rule of thumb calculations or information already on nav log if able.
 - If have time or circling, use plotter and E6B to calculate diversion information.
 - Amend flight plan if required.

Basic Instrument Maneuvers

- Review Basic Instrument Attitude flying.
 - Ensure student is able to maintain/complete:
 - Straight and level flight
 - Constant airspeed climbs
 - Constant speed and gliding descents
 - Turns to headings
 - Recovery from unusual attitudes
 - Navigation via VOR and GPS systems.

Postflight Procedures

- Ensure that the student conducts a thorough postflight inspection.

Completion Standards

This lesson is complete when the applicant can:

1. Before the flight:
 - a. Complete a navigation log with minimal errors.
 - b. Complete the climb, cruise, and descent performance calculations for the planned flight.
 - c. Obtain a weather briefing and explain how the existing weather, NOTAMs and conditions will affect the flight.
2. Without instructor guidance:
 - a. Conduct all normal and emergency checklist procedures.
3. With instructor guidance:
 - a. Intercept the planned course by using an on-course check point.

- b. Have predetermined ETAs accurate to ± 7 minutes with instructor guidance.
- c. Calculate groundspeed ± 10 knots based on actual time between check points.
- d. Maintains altitude ± 200 ft from desired altitude and ± 15 degrees from desired heading.
- e. Use radio navigation, pilotage, and dead reckoning procedures.